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Week 2026
10 – 14 March 2026, New Delhi

How Spatial Grid's AI, GIS & Immersive Spatial Models are Transforming Energy Infrastructure

Spatial Intelligence for Smart Utilities: Immersive Digital Twins for Monitoring, Training and Simulation

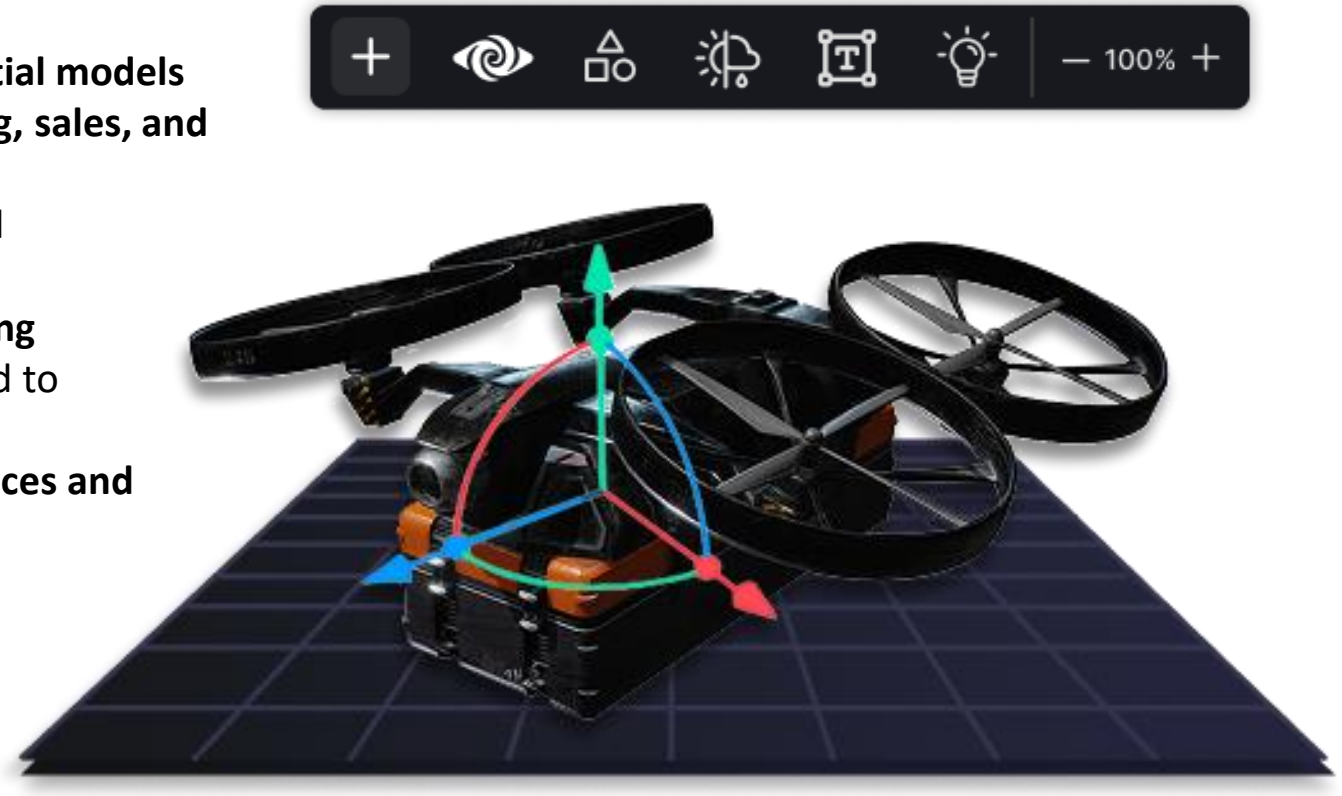
Presented By

Arjun, Founder & CXO, Spatial Grid

Spatial Grid – Building the Spatial Intelligence Layer for Infrastructure

Spatial Grid is an enterprise-grade Spatial Computing and Intelligence platform, offering cutting-edge extended reality (AR/VR/MR/Simulation) technologies emboldened by Spatial AI.

- Transform real-world environments into **interactive spatial models**
- Enable applications across **training, simulation, planning, sales, and operational intelligence**
- Combine **immersive technologies with AI-driven spatial intelligence**
- Simplify and accelerate the adoption of **spatial computing**
- Deliver **industry-specific immersive experiences** tailored to enterprise needs
- Act as the **operating system layer for mixed-reality devices and immersive platforms**



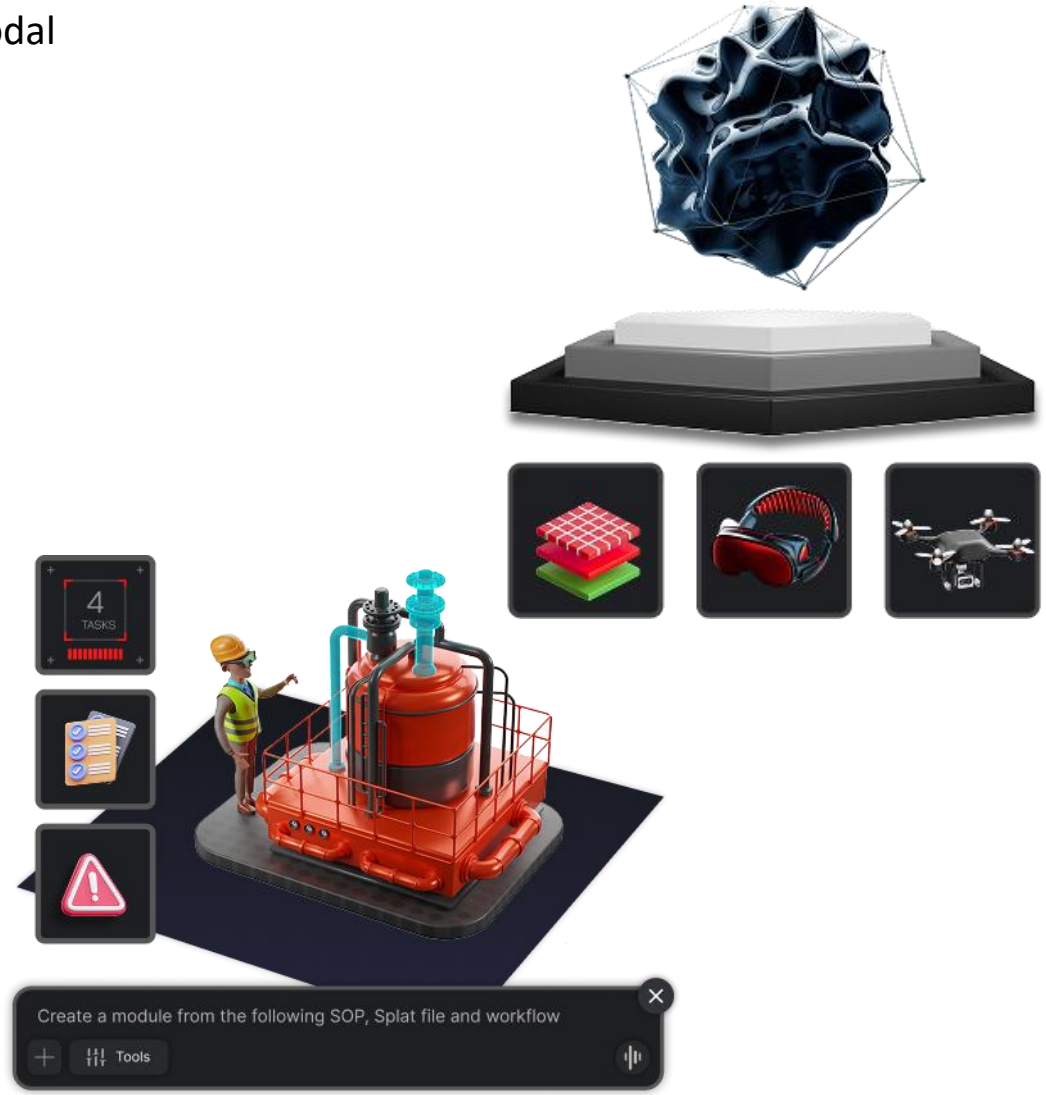
The Foundation: Immersive Spatial Models (ISMs)

Spatial Grid is built on Immersive Spatial Models (ISMs)—advanced multimodal AI systems that replicate real-world spaces and human behavior to power immersive experiences across simulations, extended-reality, and GIS systems

These AI models power immersive experiences across:

- Simulations & Simulators
- Extended Reality (AR / VR / MR)
- GIS-integrated spatial systems

ISMs act as a living digital layer over physical infrastructure—continuously enabling insight, training, and decision-making.



The Transformation of Energy and Utility Infrastructure

Utilities worldwide are facing unprecedented transformation due to:

- Renewable energy integration
- Smart grids and distributed energy networks
- Increasing infrastructure complexity
- Climate resilience and disaster management
- Workforce skill gaps in complex environments

Why Spatial Intelligence and Immersive Simulation Matter

Utilities manage highly complex physical environments including:

- Power plants
- Transmission networks
- Substations
- Renewable energy farms
- Urban energy infrastructure

Spatial Intelligence enables organizations to:

Visualize Infrastructure
Understand systems in realistic 3D environments integrated with GIS data.

Simulate Scenarios
AI-driven compute is increasing energy demand, requiring smarter, AI-optimized grids.
Test grid failures, weather events, and operational responses.

Train Workforce
Provide immersive VR training for operators and technicians.

Improve Planning
Model future infrastructure expansion and energy demand.

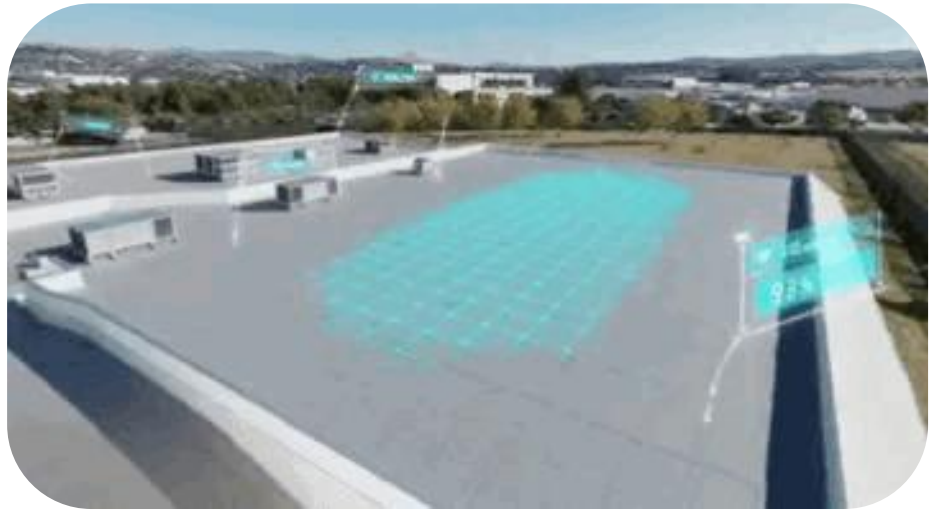
This shift moves utilities from reactive infrastructure management to predictive and simulation-driven planning.

Spatial AI for Smart Utilities

Spatial Grid enables utilities to create digital twins of infrastructure systems.

Using GIS layers, AI models, satellite data and simulation engines the platform creates immersive spatial replicas of:

- Power grids
- Renewable energy installations
- Substations and control centers
- Transmission corridors
- Urban energy systems



Key capabilities include:

Infrastructure Visualization
Real-time spatial representation of critical assets.

Digital Twin Monitoring
Operational insights through connected spatial models.

AI-assisted Analysis
Predict infrastructure stress and performance issues. This enables utilities to understand and manage infrastructure in a far more intuitive way.

Immersive Simulation for Workforce Training and Operations

A major challenge in utilities is preparing workforce for complex operations.

A major challenge in utilities is preparing workforce for complex operations. Spatial Grid provides **immersive VR and simulation environments** where workers can safely interact with digital replicas of infrastructure.

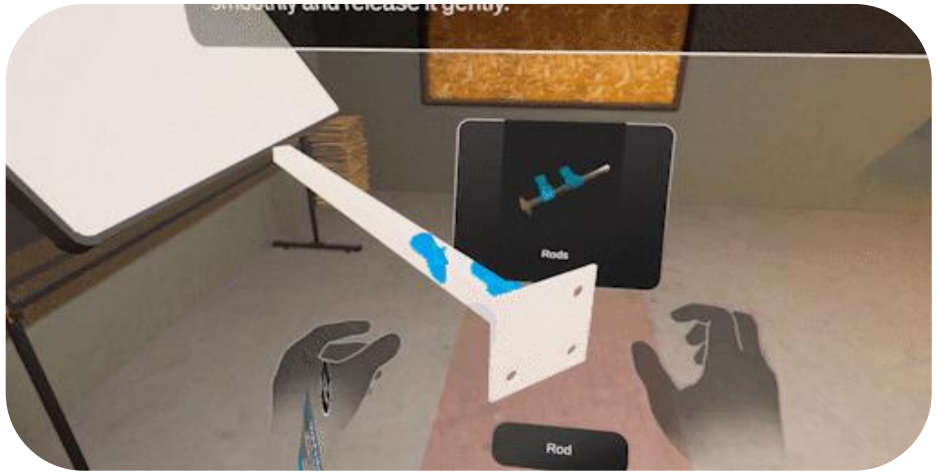
Training modules can include:

- Substation maintenance procedures
- Renewable energy system installation
- Hazard identification and safety training
- Disaster response simulations
- Control room operation training

Benefits include:

- Reduced training costs
- Improved workforce preparedness
- Standardized procedures
- Increased safety in high-risk environments

Immersive training bridges the gap between theory and real-world operations.



Use Case: VR Training for Solar Panel Installation

Objective: Improve installer training for rooftop solar deployments. Reduce safety risks during installation. Standardize installation procedures across teams

The VR Training Experience Included:

Rooftop Site Analysis

Understanding roof orientation, sunlight path, and shading analysis.

Safety Protocol Training

Proper use of helmets, harnesses, gloves, and rooftop safety procedures.

Solar Mounting Structure Installation

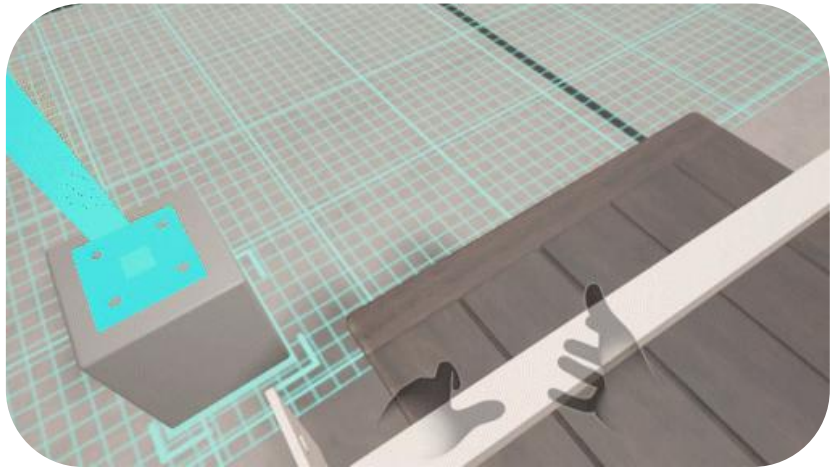
Learning how to install stands, rods, bolts, and mounting frames.

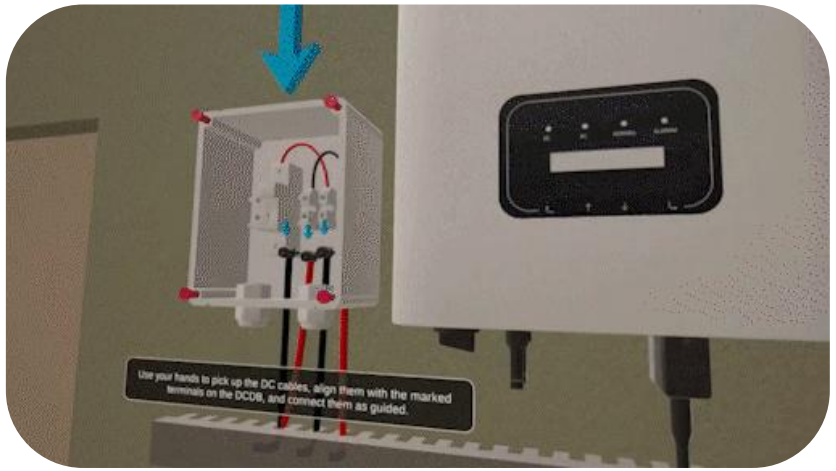
Panel Placement and Alignment

Correct positioning and angle adjustment for optimal energy generation.

Electrical Connection Simulation

Safe wiring and connection to the power system.





Key Features

- Interactive **gesture-based controls**
- Realistic rooftop simulation environment
- Step-by-step guided instructions
- Performance assessment and scoring

Impact

- Faster technician onboarding
- Improved installation accuracy
- Reduced safety incidents
- Scalable training for the growing solar workforce

In Retrospect: The future of utilities is Intelligent & Immersive

The Future of Utilities is Spatial, Intelligent and Immersive

Energy infrastructure is becoming increasingly complex & interconnected.

Utilities must adopt technologies that allow them to:

- Visualize infrastructure in real-world spatial environments
- Simulate operations before deploying them in the field
- Train workforce safely and efficiently
- Predict infrastructure risks and optimize operations
- Spatial Grid enables organizations to build digital twins of energy systems, integrate GIS data, and deploy immersive simulations for workforce training and operational planning.
- The utilities that adopt spatial intelligence platforms today will lead the smart energy ecosystems of tomorrow.

AI is rapidly increasing energy demand. As power generation and distribution grow more complex, smarter grids are needed—where AI helps optimize supply, predict demand, and improve overall grid efficiency.

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For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)